Integrating Cardiac Output

Personal:

When I was born, the pediatric doctors noticed that I had an irregular heartbeat. After further testing during the first three years of my life, I was diagnosed with a heart murmur along with a heart defect. The combination of these two heart conditions have never affected my ability to play sports or do other activities, but I have always wondered how they make me different from people with normal hearts. Since I have an irregular heartbeat, I wondered how that might affect my body physiologically compared to other people. I began to investigate cardiac output, which is the amount of blood a person's heart pumps on a given time interval. *Equation 1* is used to find cardiac output.

$$Equation\ 1\text{:}\ Cardiac\ Output\ (CO)(\frac{mL}{s}) = Stroke\ Volume\ (SV)(\frac{mL}{beat})\times Heartrate(\frac{beats}{s})$$

Aim:

Cardiac output though, is the amount of blood during a given time interval that the heart pumps, but what I really wanted to know, is how much blood my heart pumped per second on average. To find this, I had to use the techniques of integration.

Plan:

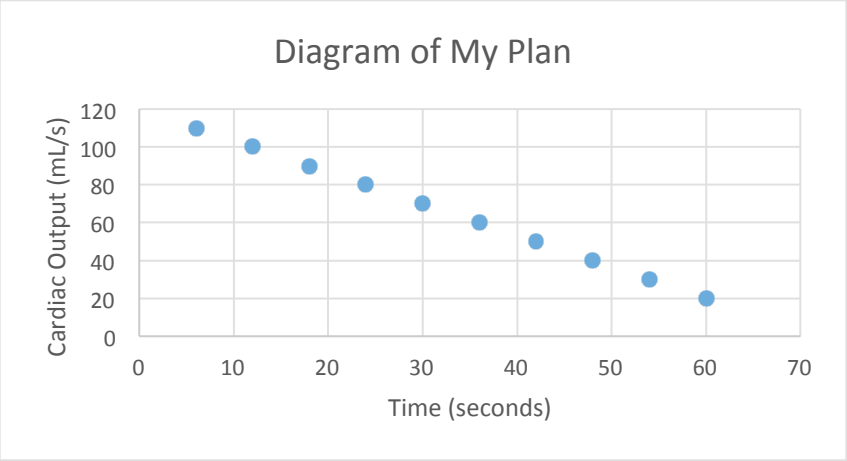
I figured that if I graphed my cardiac output, making time in seconds my x-values and the cardiac output in milliliters per second my y-values, that I could find the average value of the function. The average value of the function in this case would be the amount of blood in milliliters that my heart pumps per second on average. To find the average value, I would have to

find the area under the curve of the graph and then divide that by the change of my limits of integration, as seen by *Equation* 2.

$$Equation\ 2: Average\ Value\ of\ a\ Function = \frac{1}{x_2 - x_1} \times \int_{x_1}^{x_2} f(t)dt$$

Now that I can find the amount of blood my heart pumps per second on average, my goal is to compare my average cardiac output to a normal person's.

Before I can find the average value of my cardiac output function, I first need to devise how this graph is going to be set up. I decided that my x-values would be based on time in seconds and my y-values would be based on cardiac output in milliliters per second. Now that I have the setup of the graph outlined, I needed to figure out how I was going to create the function. Because cardiac output is based on heartrate, according to *Equation 1*, I decided that I could find my heartrate over the course of a minute. I wanted to have a total of ten points on my function, so I divided sixty seconds by ten to get six second intervals between my ten points on my graph of cardiac output. I devised that I could find the cardiac output at each of these ten points. Now, my plan for my graph is complete, but I still need to find the points to build my function.



My Process:

I now needed to find the cardiac output value at each of the ten points. According to *Equation 1*, I needed to calculate my heartrate in beats per second and my stroke volume in milliliters per beat. I first started with finding my heartrate over the course of a minute. Because I wanted ten points, I decided that I would record the amount of heartbeats I had at six, twelve, eighteen, twenty-four, thirty, thirty-six, forty-two, forty-eight, fifty-four, and sixty seconds. The amount of beats at these times is listed in *Table 1*.

Time (s)	Beats
6	8
12	14
18	20
24	24
30	29
36	34
42	39
48	44
54	48
60	53

Table 1: My heartbeat at the 10 intervals of time I desire for my graph.

Now that I have my heartbeat values at my desired time points, I can calculate my heartrate. I want my heartrate because it is a crucial part to finding my cardiac output. To find my heartrate, I used *Equation 3*, which is below.

$$Equation\ 3: Heartrate\ \left(\frac{beats}{s}\right) = \frac{Heartbeats\ (beats)}{Time\ (s)}$$

My heartrates at each of the following times are in *Table 2* below.

Time (s)	Heartrate (beats/s)
6	1.333
12	1.167
18	1.111
24	1.000
30	0.967
36	0.944
42	0.929
48	0.917
54	0.889
60	0.883

Table 2: My heartrate at each of the desired times that I will use to calculate my cardiac output.

Analysis:

Now that I have my heartrate at each particular time in seconds, I need to find my stroke volume in milliliters per beat. Upon consultation with my cardiologist, my stroke volume is not affected by either of my two heart conditions, so it is not a deciding factor in comparing the amount of blood my heart pumps per second on average to a normal person's. The calculation of stroke volume is also complex in nature. Due to these two conclusions, I decided to apply the average stroke volume, which is seventy milliliters per beat, to myself for this investigation.

Now that I have both parts of *Equation 1*, I can calculate my cardiac output at each of the particular times. These values can be seen in *Table 3* below.

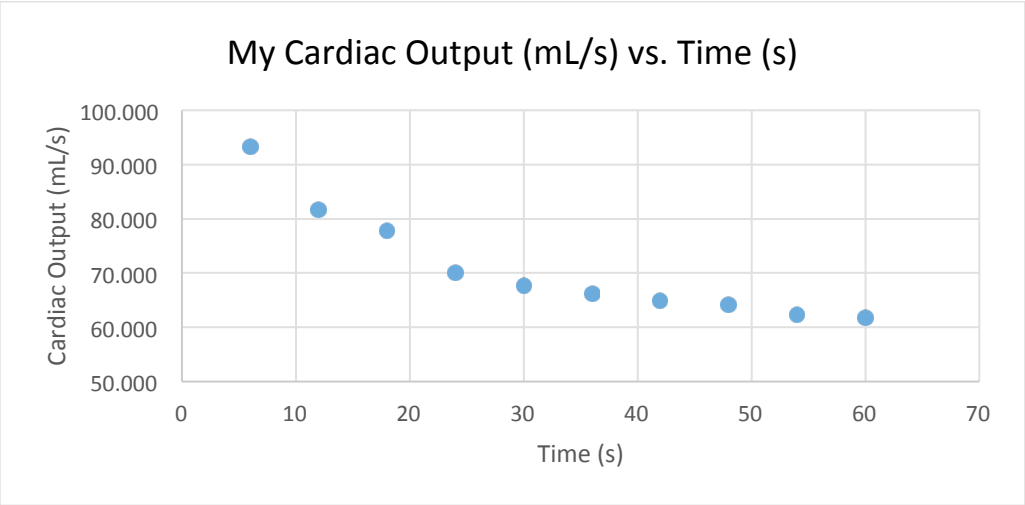
Time (s)	Cardiac Output (mL/s)
6	93.333
12	81.667
18	77.778
24	70.000
30	67.667
36	66.111
42	65.000
48	64.167
54	62.222
60	61.833

Table 3: My cardiac output at my desired time intervals.

Table 3 contains the ten points I am going to use to build my cardiac output function, with time in seconds for my x-values and cardiac output in milliliters per second for my y-values. These values are plotted in *Graph 1* below.

Reflection:

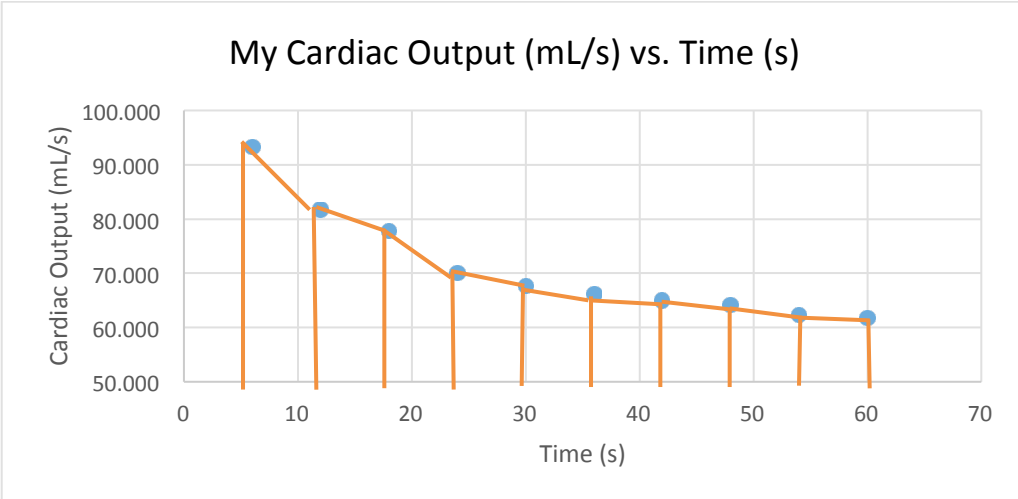
Graph 1



Graph 1 is also defined as f(x)

Analysis:

Now, I need to apply *Equation 2* to find the average value of the function. Because the function is not defined by a particular expression, I cannot simply evaluate the integral. Because of this, I have decided to use the trapezoid rule to find the average value of *Graph 1*. To visually present the trapezoid rule, I have adapted *Graph 1* to now be *Graph 1a* below.



Graph 1a

The equation version of trapezoid rule is in *Equation 4* below.

$$Equation\ 4\text{:}\ \int_{x_1}^{x_2} f(t)dt \approx \frac{1}{2}\times h\times[f(b_1)+f(b_2)]$$

Equation 4 applies if one trapezoid is used to evaluate the area under the curve. In the case of *Graph 1a*, there are nine trapezoids, so *Equation 4* will be adapted to fit this different number of trapezoids. Combining *Equation 2* and *Equation 4*, I will be able to calculate the average value of the function. The work for this calculation is provided in *Mathematical Process 1* below.

Mathematical Process 1

$$Average\ Value\ of\ Function$$

$$= \frac{1}{60-6}\times[\frac{1}{2}\times 6\times(f(6)+f(12)+f(12)+f(18)+f(18)+f(24)+f(24)$$

$$+f(30)+f(30)+f(36)+f(36)+f(42)+f(42)+f(48)+f(48)$$

$$+f(54)+f(54)+f(60))$$

$$Average\ Value\ of\ Function$$

$$= \frac{1}{54} \times [3 \times (f(6) + 2f(12) + 2f(18) + 2f(24) + 2f(30) + 2f(36) + 2f(42)$$

$$+ 2f(48) + 2f(54) + f(60))$$

$$Average\ Value\ of\ Function$$

$$= \frac{1}{54} \times [3 \times (93.333 + 2 \times 81.667 + 2 \times 77.778 + 2 \times 70.000 + 2 \times 67.667$$

$$+ 2 \times 66.111 + 2 \times 65.000 + 2 \times 64.167 + 2 \times 62.222 + 61.833)$$

$$Average\ Value\ of\ Function = 70.24\ mL/s$$

Mathematical Process 1 has determined that the average value of the function, *Graph 1*, is 70.24 milliliters per second. This means that my heart pumps 70.24 milliliters of blood per second on average. Now that I have the average value of the function for my cardiac output, I will now calculate the average value of function for a normal person's cardiac output.

Reflection:

For the graph of a normal person's cardiac output, I will use the same axes units and the same x-values as in *Graph 1*. According to *Equation* 1, to calculate the cardiac output for a normal person, I need their average heartrate and average stroke volume. I found that the average heartrate is seventy beats per minute and the average stroke volume is the same as used to find my cardiac output, seventy milliliters per beat. *Table 4* below shows the breakdown of a normal person's heartrate into heartbeats according to the same time intervals that *Graph 1* is based on.

Table 4: A normal person's heart rate at my desired time intervals that I can later use to find an average person's cardiac output.

Time (s)	Beats
6	7
12	14
18	21
24	28
30	35
36	42
42	49
48	56
54	63
60	70

Analysis:

Now, using the data from *Table 4*, I can calculate the heartrate at each of the times using *Equation 3*. These calculations are shown in *Table 5* below.

Table 5: An average person's heartrate at my desired time intervals that will contribute to calculating an average person's cardiac output.

Time (s)	Heartrate (beats/s)
6	1.167
12	1.167
18	1.167
24	1.167
30	1.167
36	1.167
42	1.167
48	1.167
54	1.167
60	1.167

I now have the heartrate at each specific time and the stroke volume for those times as well. Applying *Equation 1*, I can find the cardiac output at each of the times listed. These numbers can be seen in *Table 6* below.

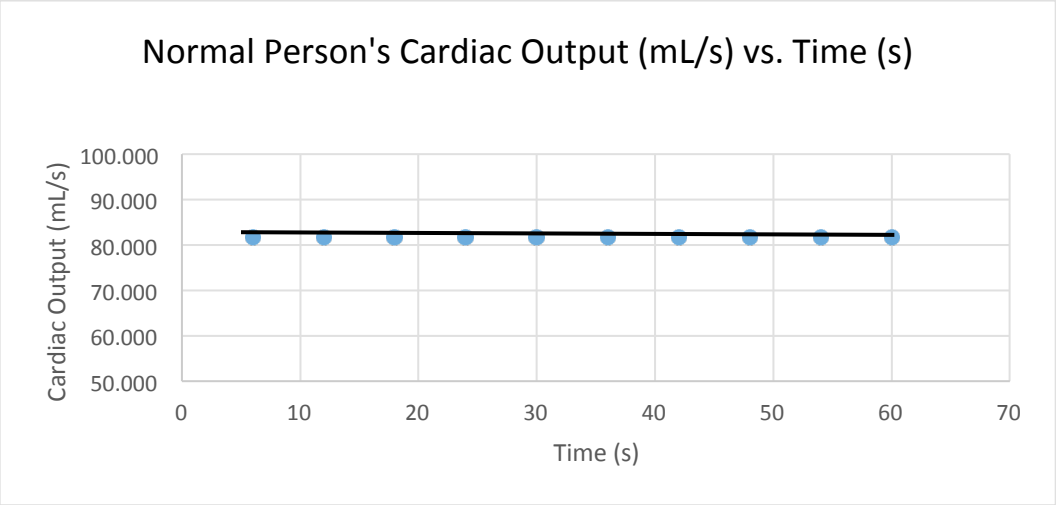
Table 6: An average person's cardiac output that I can use to create my desired function.

Time (s)	Cardiac Output (mL/s)
6	81.667
12	81.667
18	81.667
24	81.667
30	81.667
36	81.667
42	81.667
48	81.667
54	81.667
60	81.667

Using the values in *Table 6*, I can now create the graph for a normal person's cardiac output, as seen below in *Graph 2*.

Reflection:

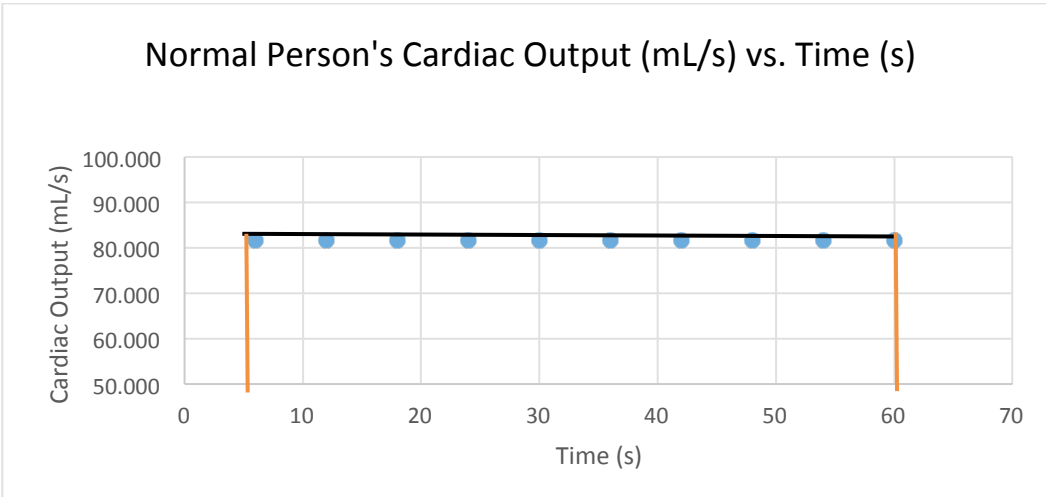
Graph 2



Graph 2 is also defined as g(x)

Analysis:

Now, I need to apply *Equation 2* to find the average value of the function. Because the function is not defined by a particular expression, I cannot simply evaluate the integral. However, due to the straight line that *Graph 2* creates, I can evaluate the area under the function based on its geometric shape, which is a rectangle, as seen in *Graph 2a* below.



Graph 2a

The equation used to find the area under the curve is in *Equation 5* below:

$$Equation\ 5\text{:}\ \int_{x_1}^{x_2} g(t)dt = h \times l$$

Now, combining *Equation 2* and *Equation 5*, I am able to calculate the average value of the *Graph 2*. The work for this calculation is provided in *Mathematical Process 2* below.

Mathematical Process 2

$$Average\ Value\ of\ Function = \frac{1}{60-6} \times [h \times l]$$

$$Average\ Value\ of\ Function = \frac{1}{54} \times [81.667 \times (60-6)$$

$$Average\ Value\ of\ Function = 81.67 \frac{mL}{s}$$

Reflection:

Mathematical Process 2 has determined that the average value of *Graph 2*, is 81.67 milliliters per second. This means that the average person's heart pumps 81.67 milliliters of blood per second on average.

Analysis:

Now that I have the average values of *Graph 1* and *Graph 2*, I can compare them. *Graph 1*'s average value was 70.24 milliliters per second while *Graph 2*'s was 81.67 milliliters per second. When looking at the two, there is 11.43 difference between the amounts of blood pumped by a normal person's heart per second on average and the amount of blood my heart pumps per second on average.

Conclusion:

My aim for this experiment was to see through this investigation if my heart conditions have an impact on my body physiologically, and if they do, what kind of impact. Due to the results of the experiment, I can conclude that there is a significant difference. My heart will pump 11.43 less milliliters of blood per second than an average human, which means that every day, my heat pumps 987,552 milliliters of blood less than a person with a normal heart. The difference here is quite significant, but what can I do with these results?

Further Reflection:

When I look at these results, I immediately think, how can I increase blood flow so that the amount of blood my heart pumps on average per second is closer to a normal person's? One way to increase blood flow is by exercising. When a human exercises, the heart pumps blood

faster through the body so that oxygen can get to the different parts of the body to keep up with the exercise. If I apply this to my findings, the amount of blood my heart pumps per second on average should increase, and this could help me regulate my blood flow. This kind of conclusion is leading me to ask more questions, such as if I did this same process but after exercising, how different would the results be? Would my number be more normal or more irregular?

Thinking about how exercise affects people's hearts, it is well known that heart rate increases and blood will flow at a faster rate. If I examine my conclusions, I begin to wonder if my cardiac output will still be lower than an average persons after exercise. Based on my knowledge of my condition and an average person, I believe that it will. If I used the same process and get my data after I exercise, for example running for 10 minutes, I could compare that to an average person's cardiac output which will lead me to the answer I desire.

Ever since I was little, I knew of my two heart conditions, but I never knew how they really affected me. This investigation has allowed me to apply my math skills and knowledge to an area of my life that is very personal. Now that I have learned how to integrate and find the average value of a function, I now know how my heart conditions affect me and to what extent they do.

Now, I begin to think about how this concept can be applied to other areas. When I think about other people's health conditions, I believe this process can be used to find how their condition makes them differ from an average person. An example of this could be looking into people who might have a lung defect. Maybe this defect could affect the volume of air that fills this person's lungs. So, this same process could be used to find on average how much air is in a person's lungs per second. Then, they could compare their numbers to a normal person's numbers like I did to see how they are different. This is just one application of the process I used

and how it can be applied to other things. I believe that my process has many more real life applications and they are yet to be discovered.

IB Mathematics SL: Exploration Feedback				Name:	
Title:	Integrating Cardiac Output	Date Set: March 24, 2017		Date Submitted: March 24, 2017	

A Communication	Score:	4	marks (out of 4 possible marks)

- Expressed ideas clearly
- Identified a clear aim for the exploration
- Focused on the aim and avoiding irrelevance
- Structured ideas in a logical manner
- Edited the exploration so that it is easy to follow
- Cited references where appropriate
- Included Introduction, rationale, description of aims, conclusion
- Graphs, tables and diagrams accompanied the work in the appropriate place and were not attached as appendices to the document.

4 The exploration is coherent, well organized, concise and complete.

Comments: Clear and concise aim, plan, reflection, analysis, and conclusion. He stays focused on the original aim throughout the paper. Used appropriate graphs and table to communicate his ideas. Mathematical processes and tables are clearly labeled and help the reader follow his exploration.

B Mathematical presentation	Score:	3	marks (out of 3 possible marks)

- Used appropriate mathematical language and representation (notation, symbols, terminology)
- Defined key terms, where required
- Selected appropriate mathematical tools (including information and communication technology)
- Expressed results to an appropriate degree of accuracy
- Used multiple forms of mathematical representation, such as formulae, diagrams, tables, charts, graphs and models, where appropriate.
- Used mathematical language when communicating mathematical ideas, reasoning and findings.
- Chose and used appropriate ICT tools such as graphic display calculators, screenshots, graphing, spreadsheets, databases, drawing and word-processing software, as appropriate, to enhance mathematical communication.

3 The mathematical presentation is appropriate throughout.

Comments: Mathematical presentation was clean and consistent. He used multiple representations to communicate his data findings and results. He defines average value and cardiac output equations. Uses appropriate terminology throughout.

C Personal engagement	Score:	4	marks (out of 4 possible marks)

- Worked independently
- Asked questions, making conjectures and investigating mathematical ideas
- Read mathematics and researched areas of interest
- Looked for and creating mathematical models for real-world situations
- Considered historical and global perspectives
- Demonstrated that he or she has "made the exploration their own"
- Expressed ideas in an individual way
- The extent to which the student engaged with the exploration and made it their own.
- Thought independently and/or creatively
- Addressed personal interest
- Presented mathematical ideas in their own way.
- Evidence that the student made the work their own, not just simply a written report.
- Asked and answered questions: "I wonder if…", "What would happen if…" "Why does that happen…"
- The student's voice could be heard

4 There is abundant evidence of outstanding personal engagement.

Comments: Student's exploration is clearly personal and meaningful (related to his heart conditions). The exploration is truly authentic. His voice can be heard throughout the paper. (ex. "Upon consultation with my cardiologist…". Student uses a model based on derivatives and rates of heartbeats to calculate blood flow. Asks creative and independent questions.

D Reflection	Score:	1	marks (out of 3 possible marks)

- How the student reviewed, analyzed and evaluated the exploration.
- Discussed the implications of results
- Considered the significance of the exploration
- Looked at possible limitations and/or extensions
- Made links to different fields and/or areas of mathematics

1 There is evidence of limited or superficial reflection.

Comments: He writes reflection above Table 1 and Table 4, but never really reflects. He discusses mathematical process more than true reflection. Some reflection occurs in the conclusion, and links to air in lungs involving the same process. He never considers the limitation of his original data of his heartbeat. Did he only take it once? What was he doing before he took it? Could taking it only once affect the reliability of his fidings.

E Use of mathematics	Score:	6	marks (out of 6 possible marks)

- Demonstrated knowledge and understanding
- Produced work that is within the level of the course. It should not be solely based on mathematics listed in the prior learning.
- Applied mathematics in different contexts
- Applied problem-solving techniques
- Recognized and explained patterns, where appropriate
- This criterion assesses to what extent students used mathematics in the exploration.
- Sophistication in mathematics may include understanding and use of challenging mathematical concepts, looking at a problem from different perspectives and seeing underlying structures to link different areas of mathematics.
- Rigour involves clarity of logic and language when making mathematical arguments and calculations.
- Precise mathematics is error-free and uses an appropriate level of accuracy at all times.

6 Relevant mathematics commensurate with the level of the course is used. The mathematics explored is correct. Thorough knowledge and understanding are demonstrated.

Comments: Demonstrates conceptual knowledge of differential equations and integral as a net change. His mathematics is accurate and error free. He makes clear logical mathematical arguments and justifies use of mathematics.

Total Marks Earned:	18	marks (out of 20 possible marks)